

Ellenbrook
CHRISTIAN COLLEGE

Year 8: MATHEMATICS 2014

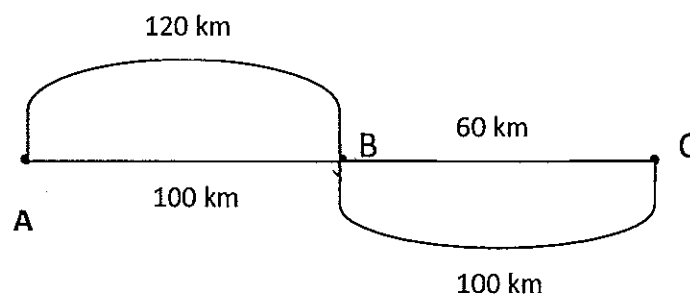
INVESTIGATION: - NETWORKS

Name	<u>Solutions.</u>
Marks	
Parent's Signature	
Comments:	

Introduction

A network is an excellent tool for situations such as finding the shortest possible distance between towns, or even planning to lay pipe lines / reticulations in your garden. In this Problem Solving task, your job is to find the shortest possible distances from one place to another and shortest possible distances to connect pipelines:

Question 1: What is the shortest distance from A to C?



Shortest Distances: A to B = 100 km

B to C = 60 km (1 mark)

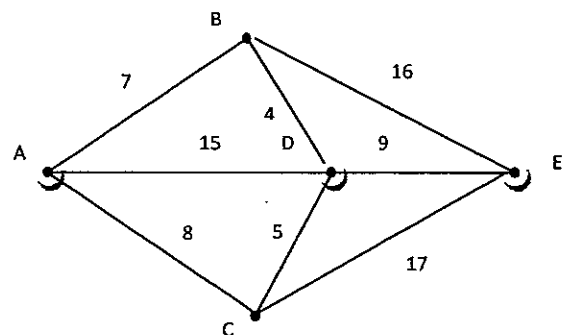
Shortest distance from A to C = 160 km (1 mark)

Question 2: What is the shortest distance from A to E? Show your working on the right.

(3 marks)

(1 mark) $7 + 4 + 9 = 20$

(1 mark) $A + B + D + E$

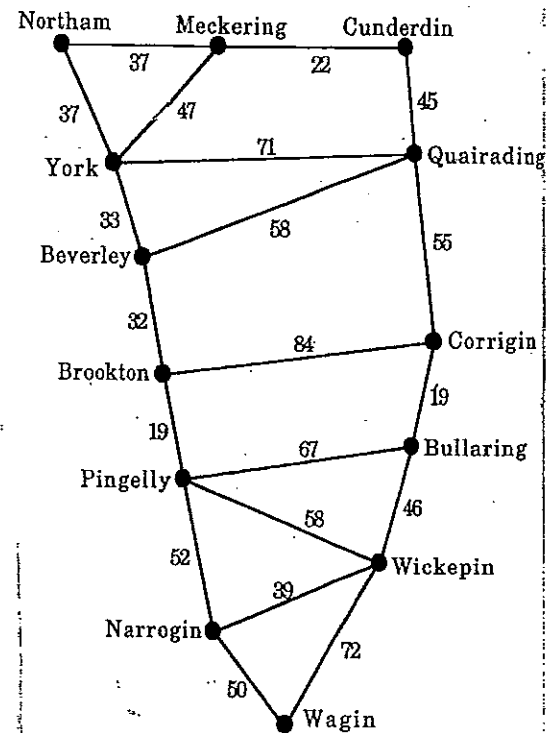


$= 20 \text{ km}.$ (If answer 3 marks)

APPLICATION Question 3: You are planning to travel from Wagin to Northam

- a. What is the shortest possible distance from Narrogin to Meckering? (3 marks)

Narrogin via York
183 km Distance Travelled.



- b. Write down all the towns that you have to pass through to get from Narrogin to Meckering at the shortest possible distance. (2 marks)

Narrogin → Pingelly → Brookton → Beverley
↓
Meckering ← York

Section 2

In this section, your task is to find out the shortest POSSIBLE CONNECTIONS to all the points for the whole system of network shown. The idea is to save money and resources

Question 4: Problem

After making a floor plan for the backyard with all the distances and all the important points for the sprinklers shown below, what is the shortest path to connect ALL the pipes to the different points in the backyard?

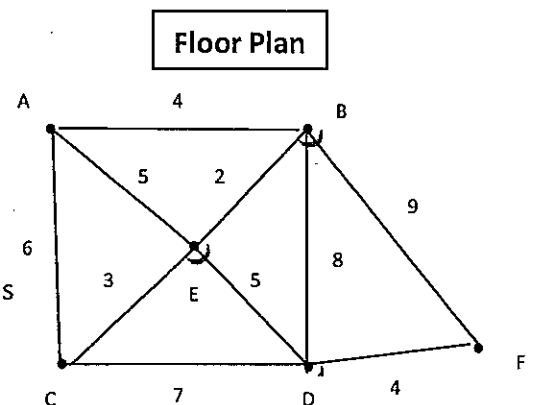
Solution:

Start from A:

Which point is shortest distance from A? B

What is this shortest distance? 4 metres

(2 marks)



From A and B:

To connect to E, the shortest distance is 2 metres from B. (2 marks)

A, B and E are connected.

From A, B and E:

To connect to C, the shortest distance is 3 metres from E. (2 marks)

A, B, E and C are connected.

From B and D:

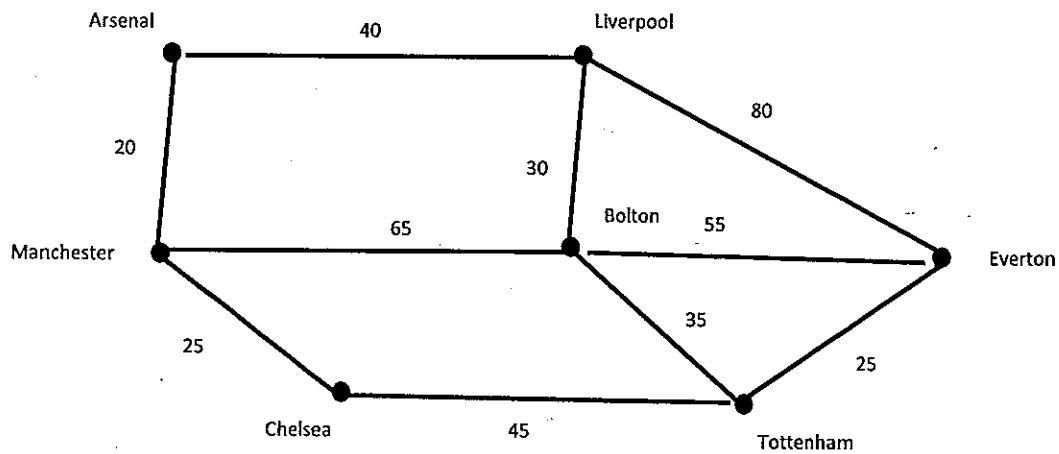
To connect to F, the shortest distance is 11 metres from F. (2 marks)

Highlight the lines that is the shortest on the diagram to check that all the points are connected (1 mark)

Total minimum length required for laying the pipes = 20 metres (1 marks)

(lose 1/2 mark if no metres or units)

Question 5: You are the town planner and you are given the task to find the shortest possible way to connect the gas pipes between the suburbs within your town council. Below is the Network collated by your surveyors.



a) Find the shortest distance for the Network. Show your working. (5 marks)

240km

M → C → T → E → B → L → A → M
 25 45 25 55 30 40 20

b) Highlight on the diagram the shortest distances connecting these towns. (1 mark)

shown above